Nonnegative and positive factorizations

Full nonnegative rank of the quadratic correlation matrix

Difficulty: extreme **Importance:** broadly interesting

Status: open; checked 2026-09-08
Area: exact NMF and lower bounds for optimization formulations

Context and notation

All factorizations are over the real numbers. For $X \in \mathbb{R}^{m\times n}_{\geq 0}$, define

$$\operatorname{rank}_+(X) = \min\{r \geq 0 : X = WH, \quad W \in \mathbb{R}^{m\times r}_{\geq 0},\ H \in \mathbb{R}^{r\times n}_{\geq 0}\}.$$

Problem statement

For each integer $n \geq 3$, let C_n be the $2^n \times 2^n$ matrix indexed by $a, b \in \{0,1\}^n$ and defined by

$$C_n(a,b) = (1 - a^\mathsf{T} b)^2.$$

Question

Is $\operatorname{rank}_+(C_n) = 2^n$ for every $n \geq 3$? All entries, including those for $a^\mathsf{T} b > 1$, are fixed by this formula.

The matrix has ordinary rank $1 + n(n+1)/2$ and is a submatrix of a slack matrix of the correlation polytope formed from valid, possibly redundant inequalities. The conjecture therefore asks for a large separation between ordinary and nonnegative matrix rank with consequences for linear programming representations.

References

Vandaele, Gillis, Glineur, and Tuyttens, *Heuristics for Exact Nonnegative Matrix Factorization*, §6.4, Conjecture 4. Gillis, *Nonnegative Matrix Factorization*, §3.7, p. 96, using the name U_n for this fixed matrix.

Status evidence

Baeckelant et al., *Computing Lower Bounds on the Nonnegative Rank via Non-Convex Optimization Solvers*, §6.7 and Appendix A.4, still states the full-rank conjecture. Searches for its resolution found none. Sergeev's July 2026 *Upper bounds for the monotone rank of the unique disjointness matrix* concerns ranks of a **partial** unique-disjointness matrix with unspecified entries when $a^\mathsf{T} b > 1$; those bounds do not determine the rank of this prescribed completion. The open $n = 3$ instance is not counted separately.